Christian Louise Asido **Module 1 : eRPT Assessment, Valuation, and Tax Declaration Engine**

1. Project Overview and Stakeholder List

Our project group is developing an automated web platform titled as Needs Assessment for a Real Property Assessment Tracking and Inter-Office Synchronization System: Resolving Disconnected Records and Workflow Delays in Municipal Assessor and Treasurer Offices in Pasacao. The main goal of our study is to conduct a needs assessment and comparative evaluation of local municipal workflows across Canaman, Magarao, San Fernando and benchmarking with Pasacao (which uses an existing eRPT setup) to solve real-world problems in local government units. Specifically, our project addresses the severe administrative frictions, valuation delays, fragmented record-keeping, and the massive operational headache of migrating decades-old legacy records dating back to 1949.

This document covers Module 1: eRPT Assessment, Valuation, and Tax Declaration Engine. This module handles property record intake, automated valuation computation, digital Tax Declaration (TD) generation, and first-stage endorsement of assessment documents. The module extends the eRPT system that Pasacao currently uses. It addresses three specific gaps found during field interviews: incomplete building-classification data fields, missing automated valuation computation for certain property types, and no online document pre-screening workflow.

Key Stakeholders

Stakeholders	Roles and Interest
Municipal Assessor	Issues the Recommending Approval for all assessment transactions before documents proceed to the Provincial Assessor. Needs accurate computation and an organized workflow queue.
Provincial Assessor's Office	Issues the final approval for all Tax Declarations in Camarines Sur. Receives endorsed assessment documents from all 35 municipal offices.
Municipal Assessor's Office Staff	Process walk-in requests daily. Requests include new Tax Declarations, transfers, reclassifications, and certifications. Staff need fast record lookup and automated valuation.
Real Property Owners and Taxpayers	Submit requests for new Tax Declarations, transfers, land holding certifications, and other assessment documents. Need fewer return trips and a clear document checklist.
Tax Mapper	Maintains tax mapping records. Updates tax maps when a property revision occurs. Needs the system to reflect updated property data after the Assessor's Office issues a Tax Declaration.
Bureau of Local Government Finance (BLGF)	National agency under the Department of Finance that oversees local government financial operations, including real property tax administration and compliance standards across all LGUs.
Capstone Team	Designs, builds, and evaluates the system against ISO/IEC 25010.

2. Information-gathering methods used and summary of findings

Field Interviews : The team conducted structured interviews with assessor staff and tax mappers at the Canaman and Pasacao Municipal Assessor's Offices using a prepared interview guide covering office setup, daily workflows, tax computation, bottlenecks, record keeping, and taxpayer experience.
Document Analysis : The team reviewed physical assessment forms and request forms such as the Certificate of Land Holding, Certified True Copy of Tax Declaration , and Transfer of Assessment form. The team also reviewed the provincial Schedule of Market Values ordinance structure as described by the interviewees.
Comparative Observation : The team compared the manual workflow of Canaman (Excel spreadsheets and paper ledgers going back to 1949) against the partially automated eRPT workflow of Pasacao.
Professor Consultation : The team held two consultation sessions with the Research Adviser and the Capstone Professor. The sessions confirmed the research scope and identified the module's primary differentiator from other GIS-based capstone groups.

Key Findings:

Canaman:

- The Canaman office has five staff members. They handle an average of three walk-in transfers per day and 15 transfers per month. Certifications are the most common daily transaction.
- Staff store property records on Excel spreadsheets and in physical ledger books. Some records date back to 1949. Retrieving a historical record takes 30 minutes to three days, depending on the record age.
- Staff compute valuation manually using the formula prescribed under Republic Act No. 7160 (Local Government Code of 1991): Area multiplied by the Unit Value from the Schedule of Market Values produces the Fair Market Value. The Fair Market Value multiplied by the applicable Assessment Level percentage produces the Assessed Value. The office has a basic computer system, but it has gaps. Certain building sub-components are not yet encodable in the current system, which causes incomplete assessment records.
- Assessment levels follow a provincial Schedule of Market Values ordinance. The ordinance covers Residential (1st, 2nd, and 3rd class), Commercial, Agricultural, and Machinery property types, each with separate unit values and assessment level percentages.
- The discount and penalty structure is standard. Taxpayers who pay by December of the current year receive a 20% discount. Taxpayers who pay in the first quarter of the following year receive a 10% discount. Late payments incur a 2% penalty per month of delay.
- The Municipal Assessor's Office issues only a Recommending Approval. The Provincial Assessor's Office in Pili issues the final approval. Municipal staff physically travel to the Provincial Capitol once a week, usually on Thursday or Friday, to transmit new documents and retrieve approved copies.
- Before the Assessor's Office issues a complete Tax Declaration for a building, it must coordinate with the Engineering Office for occupancy permits and building plans, and with the Municipal Planning and Development Office for zoning certificates when a reclassification request is involved.

Pasacao:

- The Pasacao office uses an eRPT system. The system was installed in 2022 and updated in 2024.
- The office handles 50 to 100 certification requests per day. Document verification takes about five minutes per request. Appraisal takes up to seven days for complex transactions.
- Despite using the eRPT system, staff still prepare assessments on paper first. Staff then manually encode the data into the system. The system does not allow online document submission or pre-screening. This creates a duplicate-effort workflow.

- The most common request types are: Tax Declaration, Certificate of Land Holding, Certified True Copy of Tax Declaration, Transfer and Cancellation, Certificate of No Improvement, and Effectivity Certificate.
- After staff encode property data, the eRPT system computes the assessment level and annual tax automatically. This automated computation is one of the key functions the proposed module must retain and extend.
- A known gap in the Pasacao eRPT system is that certain data fields required for accurate assessment are missing from the system's encoding forms. This causes incomplete records and requires staff to work around the system.
- No public portal exists. Taxpayers cannot check document requirements or submission status online. Staff confirmed that an online pre-screening function would reduce the number of return visits by taxpayers who arrive with incomplete documents.

Professors Consultation(Sir Chin and Sir June Areb)

- June Areb confirmed that the module should extend and improve upon the eRPT system's existing functions rather than replace it. He recommended building the module as a service layer that connects to the existing eRPT infrastructure through an application programming interface.
- Chin directed the team to address local municipal workflow efficiency before considering provincial-level bottlenecks.
- Both professors confirmed that automated Tax Declaration generation, assessment computation, and online pre-screening of requirements are valid primary functions for Module 1.

3. Functional requirements (prioritized)

MUST HAVE

- The system must accept and store property assessment request submissions. Each submission must capture the property type (Land, Building, Machinery, or Other Improvement), classification (Residential, Commercial, Agricultural, or Industrial), and Property Index Number.
- The system must allow staff to search existing property records by Owner Name, Property Index Number, Assessment Roll Number, Barangay, and Lot or Block Number. The system must return results in five seconds or less.
- The system must compute Fair Market Value automatically. The computation must follow the formula under Republic Act No. 7160, Section 198: Area multiplied by the Unit Value from the encoded Schedule of Market Values equals the Fair Market Value.
- The system must compute Assessed Value automatically. The computation must follow Republic Act No. 7160, Section 199: Fair Market Value multiplied by the applicable Assessment Level percentage equals the Assessed Value. Assessment level percentages must match the property class as the Local Government Code and the provincial Schedule of Market Values ordinance mandate.
- The system must support all standard assessment transaction types: New Assessment or Discovery, Transfer of Assessment, Revision or Re-assessment, Reclassification, Cancellation, and Subdivision or Consolidation.
- The system must generate a digital Tax Declaration. The Tax Declaration must include: full property details, computed Fair Market Value, computed Assessed Value, effective date, property classification, and the Municipal Assessor's Recommending Approval notation.
- The system must check that all required documents are present before the request moves to the appraisal or verification stage. For a Transfer of Assessment, the system requires a Deed of Absolute Sale (or equivalent conveyance document). For a new building assessment, the system requires a Building Permit and a Certificate of Occupancy.
- The system must allow staff to encode all building sub-components. Sub-components include: structural type, floor area, year built, depreciation rate, and accessories or additional improvements. This requirement addresses the encoding gap found in the Canaman field interview.

SHOULD HAVE
● The system should allow taxpayers or authorized representatives to submit initial assessment requirements online. The system should check the document checklist and flag missing items. It should notify the taxpayer of any missing documents before the taxpayer visits the office. ● The system should compute the annual Real Property Tax due automatically. The computation must follow Republic Act No. 7160, Section 233: Basic Real Property Tax equals Assessed Value multiplied by the applicable tax rate (not exceeding 1% for provinces). The computation must also include the Special Education Fund levy of 1% of the Assessed Value, applicable discounts (20% for advance payment by December of the current year; 10% for first-quarter payment of the following year), and penalties (2% per month of delay, compounded per quarter). ● The system should store the complete assessment history of each property. The history must include all prior Tax Declarations, revisions, reclassifications, and recorded owners from the earliest available declaration. Staff must be able to view this history in a property timeline panel. ● The system should generate a Notice of Assessment document automatically when a Tax Declaration is final. The system must provide this document to the taxpayer as official notification of the new assessed value. ● The system should produce a formatted Assessment Roll entry. The system must allow staff to export this entry and transmit it to the Treasurer's Office as a digital record.

COULD HAVE
● The system could allow staff to attach scanned supporting documents to the property record. Documents include: Deed of Sale, Land Title, Building Permit, and Certificate of Occupancy. ● The system could display a taxpayer-facing document checklist page. The page must show the documents required per transaction type. Account registration must not be required to view this page. ● The system could flag properties where the encoded digital record does not match the paper-based historical record. The system must prompt a reconciliation workflow for flagged properties.

WONT HAVE
● This Module will not process online tax payments or connect to payment gateways. Those functions belong to the Treasurer's Office scope (Module 2). ● The system will not include GIS or tax mapping functions. Those functions belong to a separate capstone group to avoid scope conflict. ● This Module will not replace the formal physical signing process at the Provincial Capitol in Pili. That function is the scope of Module 3.

4. Non-functional requirements (prioritized)

MUST HAVE	
Non-Functional Requirement	**Measurable Target**
Performance Efficiency : The system must return property record search results within an acceptable time. This applies to all records, including historical data.	Search response ≤ 5 seconds for any record.
Functional Suitability : Assessment computations (Fair Market Value, Assessed Value, and Real Property Tax due) must match manual verification results at all times.	0% computation error rate versus manual benchmark calculations during evaluation.

Security and Data Privacy : The system must restrict access to all stored property records and owner information. Only authenticated assessor staff may access or change assessment records. This requirement complies with Republic Act No. 10173 (Data Privacy Act of 2012).	Role-Based Access Control enforced for all record access and modification.
Usability : Assessor staff with minimal technical training must complete a standard Tax Declaration generation without external help.	ISO/IEC 25010 Usability score ≥ 4.0 out of 5.0 on the user evaluation survey.

SHOULD HAVE	
Non-Functional Requirement	**Measurable Target**
Reliability : The system must handle multiple simultaneous users without data loss or corruption. This applies during peak periods such as end-of-year payment season and transfer request rushes.	The system remains stable under at least five concurrent users.
Maintainability : An authorized administrator must update Schedule of Market Values unit values and assessment level rates without changing the source code. The province updates the Schedule of Market Values ordinance periodically.	Admin panel allows rate updates without developer intervention.
Processing Time Reduction : The system must reduce the time to complete a new assessment or Tax Declaration generation compared to the manual baseline.	Average Tax Declaration generation time ≤ 15 minutes versus the current manual baseline of 30 minutes to three days.

COULD HAVE	
Non-Functional Requirement	**Measurable Target**
Accessibility : The taxpayer-facing pre-screening portal must work on mobile browsers.	Responsive design functional on screen widths ≥ 360 pixels.

WONT HAVE
● This module will not include blockchain validation of Tax Declarations. The Capstone Professor noted this as a possible future enhancement. It is outside the scope of this academic year.

5. Key use cases or user stories (3–5 high-priority ones)

New Building Assessment (Tax Declaration Generation):
As an Assessor's Office staff member, I want to encode a new building's complete details into the system and have it compute the Fair Market Value, Assessed Value, and Real Property Tax automatically, so that I can generate an accurate digital Tax Declaration within minutes instead of preparing it manually over several hours.

Transfer of Assessment with Online Pre-Screening:
As a real property owner who needs to transfer a Tax Declaration to a new owner's name, I want to check and submit my documentary requirements online before visiting the Assessor's Office, so that I arrive with complete documents and avoid making multiple return trips.

Historical Property Record Lookup:
As an Assessor's Office staff member, I want to search for a property's complete assessment history using the owner's name or Property Index Number, so that I can verify prior assessments and retrieve records going back to 1949 within seconds instead of spending 30 minutes to three days searching physical ledgers.
Annual Real Property Tax Computation with Discounts and Penalties:
As an Assessor's Office staff member, I want the system to automatically compute a property's annual Real Property Tax due including any applicable discount or penalty based on the payment date so that I can give the taxpayer an accurate billing breakdown on demand without performing the calculation manually.

6. **Assumptions, constraints, and open issues**

ASSUMPTIONS
● The Assessor's Office will provide the Schedule of Market Values ordinance data for encoding into the valuation database. The Pasacao interview confirmed that the ordinance exists in both book and system form. ● The Municipal Assessor will participate in system evaluation and will confirm the Recommending Approval workflow representation. ● The team will use mock or masked property data during development and testing to comply with Republic Act No. 10173. The team will use actual property data only during the supervised field evaluation phase. ● Internet connectivity at the target offices is adequate for a web-based system. ● The system covers only the assessment functions of the Assessor's Office. Real Property Tax collection and payment processing remain with the Treasurer's Office (Module 2). ● During the pilot phase, the Provincial Capitol approval step remains a physical process. Module 1 generates only the Recommending Approval packet for endorsement. Module 3 handles digital approval routing.

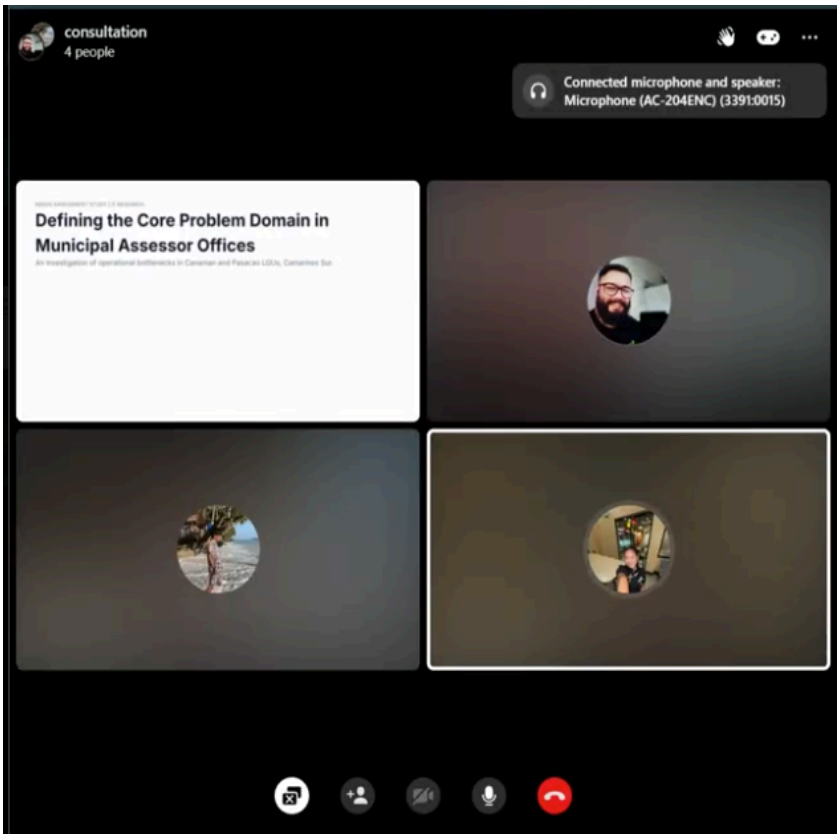
CONSTRAINTS	TYPE
Assessment levels and tax rates follow Republic Act No. 7160 (Local Government Code) and the provincial Schedule of Market Values ordinance. The system cannot change these rates without a new ordinance from the Province.	Policy
All municipal Assessors in Camarines Sur operate under the Provincial Assessor's authority. Only the Provincial Assessor issues the final approval signature on Tax Declarations. The system produces only a Recommending Approval package at the municipal level in Module 1.	Legal and Policy
The eRPT system used in the target municipalities is a locally developed and customized system. This module extends it as a service layer. The team will not change the core codebase of the existing eRPT system. The Capstone Professor recommended this approach to maintain institutional compatibility.	Technical
The academic project timeline runs from July to December 2026. The deliverable for this phase is a working prototype. The complete production system is scheduled for the following academic year's capstone continuation.	Time
The Real Property Valuation and Assessment Reform Act (Republic Act No. 12001) targets full implementation in 2028. Current assessment computations must follow the existing Schedule of Market Values framework, not the reform act framework.	Regulatory
Legacy records from 1949 to approximately 2000 in Canaman exist only in physical ledger books. Digitizing these records requires manual data entry by assessor staff. Historical retrieval in the system is limited to records that staff encode during the pilot run.	Data

OPEN ISSUES
● Will the Assessor's Office provide an exported digital dataset (such as a spreadsheet or database export) of the Schedule of Market Values ordinance, or must the development team manually encode the unit values and classification tables from the physical ordinance book into the system database? ● The system must handle requests where the person at the counter is not the property owner. Those cases require a Special Power of Attorney or a written Authorization Letter. The team must confirm the exact verification logic with the Assessor's Office. ● A Transfer of Assessment requires a Certificate Authorizing Registration from the Bureau of Internal Revenue. The system currently plans to accept a scanned upload only. Whether any real-time verification is feasible within the project timeline is not yet decided. ● The team and the Assessor's Office must agree on how long the office will run both the existing workflow and the new system at the same time. This comparison period provides data for the Paired t-test evaluation. A duration of two weeks is proposed. ● The team must confirm the exact Assessment Roll format that the Treasurer's Office expects. This ensures that the Module 1 export function produces a compatible file.

7. Attachment of Data Gathering Evidences

Sample Interview Questions
1. Can you walk us through the step-by-step process from the moment a taxpayer arrives at your service counter until they receive their finished document or clearance? 2. What specific documentary requirements must property owners submit for each of the following major transactions? a. New Tax Declaration: Requirements for newly declared land, new buildings/houses, or machinery. b. Transfer of Ownership: Requirements when transferring a property (such as Deed of Sale, BIR tax clearance, transfer tax receipt, or previous clearances). c. Reassessment of Property: Requirements when an owner requests a value correction, property reclassification, or physical inspection. d. Issuance of Clearances: Requirements for getting a Tax Clearance, Certificate of No Property, or Certificate of No Improvement. 3. How do you compute real property taxes in your office from start to finish? a. Market Value & Assessed Value: How do you determine market value, and what assessment percentages are applied based on classification (residential, agricultural, commercial)? b. Tax Rates: What tax percentages are applied to get the annual tax due (such as the 1% basic tax and 1% Special Education Fund)? c. Discounts and Penalties: What discounts are given for prompt or advance payments, and what interest or penalties are charged for late payments? 4. How do personnel search for an existing property record, tax declaration, or property identification number when a client inquires at the counter? 5. On average, how long does it take to complete one full transaction from initial queuing to document release? Roughly how many minutes are spent on: a. Requirement Checking: Checking if the client's submitted papers are complete. b. Record Search: Locating the property in physical record books, maps, or file folders. c. Value & Tax Computation: Calculating the assessed value and tax amounts. d. Approval & Release: Review, official signature, municipal seal, and releasing the document. 6. How often do calculation mistakes or clerical errors happen during manual tax computations, what usually causes them, and how are they corrected? 7. If an online portal allowed taxpayers to check requirements, submit initial papers, and see tax estimates before visiting, which tasks would this reduce or simplify for your staff?

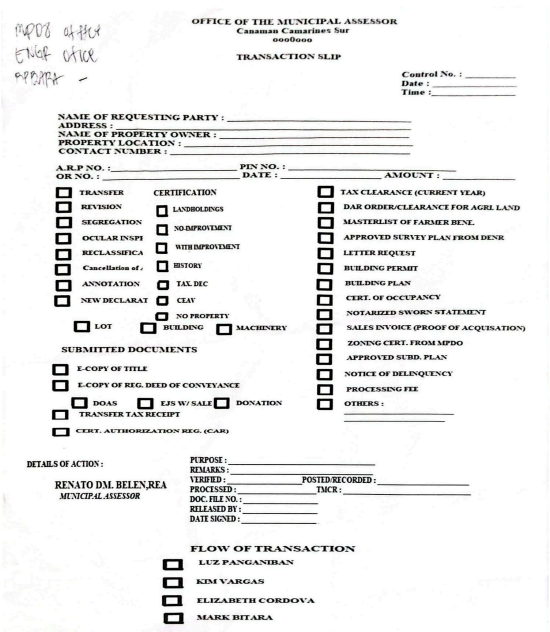
Pictures of Online consultation and Interviews







Sample Documents



OFFICE OF THE MUNICIPAL ASSESSOR
Canaman Camarines Sur
oooOooo

TRANSACTION SLIP

Control No. : _____
Date : _____
Time : _____

NAME OF REQUESTING PARTY : _____
ADDRESS : _____
NAME OF PROPERTY OWNER : _____
PROPERTY LOCATION : _____
CONTACT NUMBER : _____

A.R.P NO. : _____ PIN NO. : _____
OR NO. : _____ DATE : _____ AMOUNT : _____

- ☐ TRANSFER
- ☐ REVISION
- ☐ SEGREGATION
- ☐ OCULAR INSPI
- ☐ RECLASSIFICA
- ☐ Cancellation of /
- ☐ ANNOTATION
- ☐ NEW DECLARAT

CERTIFICATION
- ☐ LANDBOLDINGS
- ☐ NO IMPROVMENT
- ☐ WITH IMPROVEMENT
- ☐ HISTORY
- ☐ TAX. DEC
- ☐ CLAV
- ☐ NO PROPERTY

- ☐ LOT ☐ BUILDING ☐ MACHINERY

SUBMITTED DOCUMENTS
- ☐ E-COPY OF TITLE
- ☐ E-COPY OF REG. DEED OF CONVEYANCE
- ☐ DOAS ☐ EJS W/ SALE ☐ DONATION
- ☐ TRANSFER TAX RECEIPT
- ☐ CERT. AUTHORIZATION REG. (CAR)

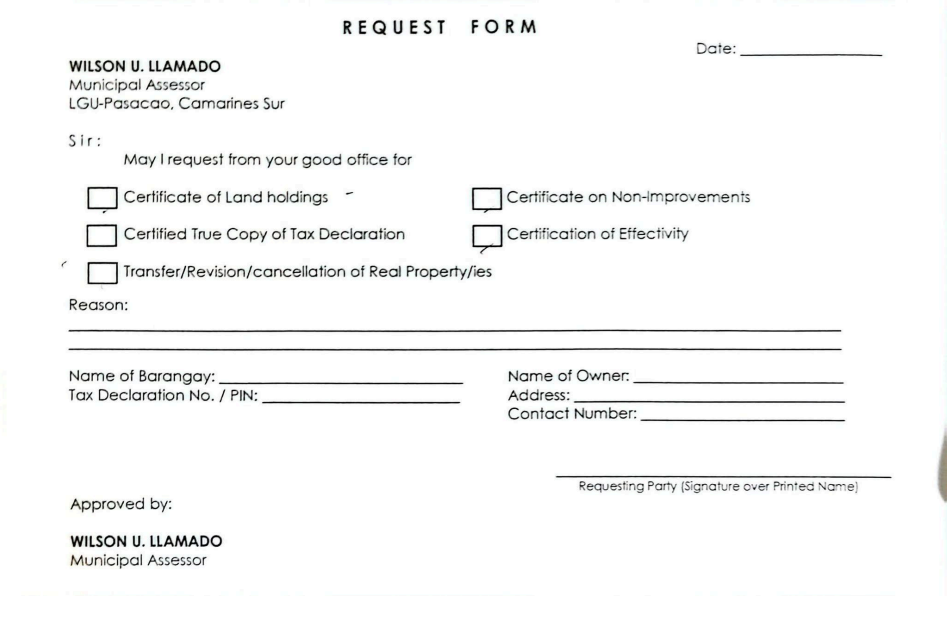
- ☐ TAX CLEARANCE (CURRENT YEAR)
- ☐ DAR ORDER/CLEARANCE FOR AGRL LAND
- ☐ MASTERLIST OF FARMER BENE.
- ☐ APPROVED SURVEY PLAN FROM DENR
- ☐ LETTER REQUEST
- ☐ BUILDING PERMIT
- ☐ BUILDING PLAN
- ☐ CERT. OF OCCUPANCY
- ☐ NOTARIZED SWORN STATEMENT
- ☐ SALES INVOICE (PROOF OF ACQUISATION)
- ☐ ZONING CERT. FROM MPDO
- ☐ APPROVED SUBD. PLAN
- ☐ NOTICE OF DELINQUENCY
- ☐ PROCESSING FEE
- ☐ OTHERS : _____

DETAILS OF ACTION :

RENATO DM. BELEN,REA
MUNICIPAL ASSESSOR

PURPOSE : _____
REMARKS : _____
VERIFIED : _____ POSTED/RECORDED : _____
PROCESSED : _____ TMCR : _____
DOC. FILE NO. : _____
RELEASED BY : _____
DATE SIGNED : _____

FLOW OF TRANSACTION
- ☐ LUZ PANGANIBAN
- ☐ KIM VARGAS
- ☐ ELIZABETH CORDOVA
- ☐ MARK BITARA



REQUEST FORM

Date: _____

WILSON U. LLAMADO
Municipal Assessor
LGU-Pasacao, Camarines Sur

Sir:

May I request from your good office for

- ☐ Certificate of Land holdings
- ☐ Certificate on Non-Improvements
- ☐ Certified True Copy of Tax Declaration
- ☐ Certification of Effectivity
- ☐ Transfer/Revision/cancellation of Real Property/ies

Reason:

_____
_____

Name of Barangay: _____
Tax Declaration No. / PIN: _____

Name of Owner: _____
Address: _____
Contact Number: _____

_____
Requesting Party (Signature over Printed Name)

Approved by:

WILSON U. LLAMADO
Municipal Assessor